LOVE KUSH PRANU

New York, NY | (929) 414-7472 | lp2989@nyu.edu
Linkedin | Github | Portfolio

EDUCATION

New York University, Tandon School of Engineering | New York, NY
Master of Science in Cybersecurity, GPA 3.9/4.0 | Expected December 2026

- Relevant coursework: AI in Cybersecurity, Application Security, Applied Cryptography, Information Systems Security Engineering, Malware Analysis.

National Institute of Technology, Calicut | India
Bachelor of Technology in Electrical and Electronics Engineering, CGPA 8.2/10 | 2023

EXPERIENCE

NYU OSIRIS Lab | New York, NY — September 2024 - Present
Research Coordinator and Infrastructure Lead

- Designed and operated a serverless AWS architecture (API Gateway, Lambda, DynamoDB, EventBridge, EC2) with infrastructure-as-code and EC2 lifecycle automation, running a CTF platform for 5,000+ participants with least-privilege IAM.
- Built a Slurm-backed GPU grading pipeline that automated evaluation and scoring across distributed compute nodes, cutting manual grading effort for the competition.
- Coordinated cross-functional research tracks and authored reverse-engineering challenge software for CSAW CTF, a large student-run cybersecurity competition.

Pentadots Solutions | London, UK (Remote) — September 2023 - January 2025
Cybersecurity Engineer

- Wrote proof-of-concept exploit code and automation across five or more production deployments, chaining candidate weaknesses into reproducible impact and authoring the remediation documentation.
- Built zero-trust frameworks with microsegmentation and least-privilege access controls, improving defense efficiency by 30%.

Robert Bosch India Ltd. | Coimbatore, India — June 2022 - August 2022
E-Drive System Security Intern

- Analyzed embedded and networking attack surfaces (CAN bus, ECU, firmware) and ran threat analysis and risk assessment (TARA) with Porsche engineering teams against ISO 26262, raising validation coverage by 25%.

PROJECTS

Orion, AI Vulnerability Scanner | Python, LLM Agents, MCP, Neo4j, Joern

- Built a multi-agent AI system that orchestrates concurrent large language model (LLM) discovery agents over a Joern code property graph in Neo4j, exposing read-only tools through the Model Context Protocol (MCP) and pairing each lead with an isolated verifier agent that filters false positives.
- Engineered clean, maintainable, framework-agnostic code with roughly 26 unit tests; reached 14 of 15 detections at zero false positives on NodeGoat and found a four-file shell-injection remote code execution (RCE) in a 160,000-node production TypeScript repository, with proof-of-concept exploits.

Gemma Intrusion Detection, LLM Fine-Tuning Pipeline | Python, PyTorch, Unsloth, QLoRA

- Built an end-to-end machine learning pipeline that fine-tunes Google's Gemma LLM on the UNSW-NB15 dataset for network intrusion detection, using Alpaca-format data and memory-safe QLoRA on consumer GPU hardware.

RaceFuzz, Linux Kernel Race Condition Fuzzer | Go, C

- Built a distributed systems fuzzer for Linux TCP reassembly with a Go host orchestrator and a C kernel module injecting packets on nanosecond hrtimers via generic netlink; authored 16 race templates across BPF, nftables, and net/sched.

SKILLS

Certifications and CTF: CompTIA Security+ and CySA+, CCNA (in progress); NYUSec CTF team (12 US, 87 global).
Security and Networking: Application security, secure code review, static analysis (Joern CPG), threat modeling (STRIDE, TARA), cryptography; TCP/IP fundamentals, netlink,SCRUM, eBPF/BPF, embedded systems (CAN bus, ECU, firmware)
Programming: Python (advanced), C/C++, Go, Bash, Perl, SQL, Cypher.
AI/ML: Machine Learning, large language models (LLMs), LLM fine-tuning (Unsloth, QLoRA), PyTorch, Gemma, multi-agent systems, Model Context Protocol (MCP).
Cloud, Systems and DevOps: AWS (Lambda, API Gateway, DynamoDB, EventBridge, EC2, IAM, VPC), serverless, infrastructure-as-code, lifecycle automation, Slurm, distributed systems, Linux kernel internals, Git